Robust and Compact Recognition of Code-Switched Speech

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Abstract

In English–Mandarin code-switched speech, the words from the other language (*embedded* words) are often the ones that carry the meaning, yet speech recognisers are judged by one overall error rate in which these words are a small share. We measure Whisper's errors by the role each word plays in its sentence, on two corpora whose embedded language is reversed (ASCEND and NSC Part 4), and ask what fine-tuning, new training and decoding methods, and quantization for on-device use do to embedded words. The same word is more often wrong when it is embedded; fine-tuning removes this penalty for embedded Mandarin but not for embedded English, and the remaining errors are rarely translations; many are sound-alike transliterations, though a small human audit suggests our automatic labels overcount them. A role-weighted loss combined with a semantic minimum-risk loss lowers both overall and embedded-word error on NSC, and language-ID re-ranking helps when it is applied only to clips that contain a switch. 8-bit quantization is safe, but at 4-bit the embedded error of the tiny and base models rises 1.4 to 4.4 times as much as matrix error. Keeping sensitive layers at 16-bit or QLoRA fine-tuning recovers most of this loss in NF4, but not in the 4-bit format used on device. On 120 phone recordings by three speakers, 8-bit again changes nothing, while 4-bit raises embedded error by 7 to 9 points for base and, in NF4, even for small, with no measurable change in matrix error.

1 Introduction

Many Singaporeans switch between English and Mandarin within a single sentence. In "我明天要交 assignment 啦" ("I have to submit my assignment tomorrow"), the sentence is mainly Mandarin, its *matrix* language, and the English word "assignment" is *embedded* in it. Such code-switched speech is hard for speech recognisers, and the embedded words are often the ones that matter most: a drug name, a deadline, a course code or a place.

Settings such as clinics, counselling and live captions for deaf users often cannot send audio to the cloud, so the recogniser has to run on the user's own device. The usual way to make a model small enough is *quantization*: storing each weight in 8 or 4 bits instead of 16. Quantized models are normally judged by one overall error rate. In code-switched speech, however, embedded words are a small share of all words, so they can fail badly while the overall error rate barely moves.

Fine-tuning Whisper (Radford et al., 2023) on code-switched Mandarin–English speech improves accuracy (Yang et al., 2024; Zhou et al., 2025), and Ugan et al. (2025) give switch points extra weight during training. Quantization of Whisper has been studied on monolingual speech (Söhler et al., 2026; Ferraz, 2024), where it hurts lower-resource languages more (Ferraz, 2024). To our knowledge, no prior work measures how quantization changes errors on embedded words separately from errors on the rest of a mixed sentence, or asks whether embedded words fail because of the language they are in or because of the role they play in the sentence.

We study these questions with Whisper tiny, base and small on two corpora in which the embedded language is reversed: ASCEND (Lovenia et al., 2022), mostly Mandarin with English embedded, and the Mandarin-speaker part of NSC Part 4 (Koh et al., 2019), mostly English with Mandarin embedded. We split every sentence into units (one Chinese character or one English word) and label each unit as

embedded (E), matrix next to a switch (M_s) or matrix elsewhere (M_i), so that every error can be charged to a role. We ask three research questions: **RQ1**: where does Whisper fail on code-switched speech, and is it because of a unit's role or its language? **RQ2**: do training and decoding methods aimed at embedded units reduce these errors at 16-bit? **RQ3**: does quantization hurt embedded units more than matrix units, and can the damage be repaired? Our hypotheses were that embedded units fail partly because of their role and not only their language (H1), that training and decoding aimed at embedded units lower their error without raising overall error (H2), and that quantization hurts embedded units more than matrix units, most in the smallest models (H3).

Our contributions and main findings are:

- **A role-level error analysis** (RQ1). The same word has a higher error rate when it is embedded than when it belongs to the matrix language, even after controlling for language, word frequency and length. Fine-tuning removes this penalty for embedded Mandarin but not for embedded English. Contrary to our hypothesis, embedded words are rarely replaced by their translation; many are instead *transliterated* into the other script ("deadline"→"得来"), although a small human audit suggests our automatic labels overcount transliterations.

- **Training and decoding methods for embedded units** (RQ2). A role-weighted loss combined with a sentence-level semantic loss lowers both overall and embedded-unit error over plain fine-tuning on both NSC test sets. Re-ranking candidate transcripts with a language-ID head helps embedded units in Mandarin-main speech but hurts English-main speech; running it only on clips where the head detects a language switch keeps the gain and removes the harm.

- **A measure of quantization fragility for embedded units and two fixes** (RQ3). 8-bit quantization is safe in every backend we test. At 4-bit, embedded-unit error rises 1.4 to 4.4 times as much as matrix error for the tiny and base models, while small is robust on the corpora; on our own phone recordings, NF4 hurts small's embedded words too. Keeping the most sensitive layers at 16-bit, or quantization-aware fine-tuning with QLoRA (Dettmers et al., 2023), recovers about half to nine-tenths of the embedded-unit loss on most affected slices, but the QLoRA gain does not carry over to the 4-bit format used on device.

Our best system, fine-tuned Whisper small with both losses, lowers the overall error on the NSC test set from 55.1% to 20.6% and the embedded-unit error from 76.7% to 21.3% compared with the off-the-shelf model. Stored in 8-bit with whisper.cpp (Gerganov et al.), it is a 252 MB file that runs at 0.57 times real time on a laptop CPU. Code, scoring tools and all result tables are at `https://github.com/nibess1/cs425_codeswitch`.

2 Related Work

Whisper on code-switched speech. Whisper (Radford et al., 2023), trained on weakly labelled multilingual speech, is a strong starting point for Mandarin–English recognition. Fine-tuning it on code-switched corpora lowers its error rate considerably (Yang et al., 2024; Zhou et al., 2025). Even at full precision, Whisper can drop embedded words or output their translation instead of a transcription (Peng et al., 2023). These studies report one overall error rate, so they do not show which words in a mixed sentence are lost, or whether the losses come from a word's language or from its role in the sentence.

Training for switches and for sequence-level cost. Ugan et al. (2025) give tokens at switch points extra weight during training, the closest prior work to our role-weighted loss; we weight embedded units and matrix units next to a switch separately. Minimum word error rate training (Prabhavalkar et al., 2018) lowers the expected error over a set of candidate transcripts instead of the error of each token; our semantic loss follows this approach with a cost that gives partial credit to correct translations,

found with a dictionary (MDBG) and LaBSE sentence embeddings (Feng et al., 2022).

Quantizing speech models. Weights can be stored in 8 bits (Dettmers et al., 2022) or in 4 bits with NF4, and low-rank adapters (Hu et al., 2022) can be trained on top of a 4-bit model (Dettmers et al., 2023). For Whisper, Söhler et al. (2026) study how quantization design choices affect recognition error, and Ferraz (2024) find that compression hurts languages with less training data more. On-device runtimes such as CTranslate2 (OpenNMT) and whisper.cpp (Gerganov et al.) use their own quantization formats. These studies do not look at code-switched speech, where a small group of embedded words could be damaged while the overall error rate barely changes.

What we change. Our starting point is Whisper fully fine-tuned on code-switched Mandarin–English speech, as in Yang et al. (2024) and Zhou et al. (2025). Relative to that work we (i) score every error by the role of the unit it hits instead of reporting one overall rate; (ii) weight embedded units and matrix units at a switch separately in the loss, where Ugan et al. (2025) weight switch points only; (iii) add a minimum-risk loss whose cost forgives correct translations, where MWER training (Prabhavalkar et al., 2018) counts every word alike; (iv) re-rank candidates with a frame-level language-ID head, applied only when a switch is heard; and (v) measure quantization separately for embedded and matrix units with a fragility ratio, in the formats used on device, and test two repairs.

3 Task and Metrics

Units and roles. A shared text normaliser turns every reference and hypothesis into *units*: one Chinese character, one English word or one digit string. The *matrix* language of a sentence is the language with more words once the Chinese part is segmented into words; sentences with a tie are *balanced* and count towards the overall error rate only. Every unit gets one of three roles that do not overlap: *embedded* (E), a unit in the non-matrix language; *matrix at a switch* ($\mathrm{M_s}$), a matrix unit directly before or after a change of language; and *matrix elsewhere* ($\mathrm{M_i}$). In "我明天要交 assignment 啦", "assignment" is E, 交 and 啦 are $\mathrm{M_s}$ and 我明天要 are $\mathrm{M_i}$.

Error rates. The mixed error rate is

$$\mathrm{MER} = \frac{S + D + I}{N}, \tag{1}$$

where S, D and I are the numbers of substituted, deleted and inserted units in the best alignment of hypothesis and reference, and N is the number of reference units. To give each error a role, a second alignment pass charges every block of errors, including its insertions, to the reference units it covers. The error rate of a role is then the errors charged to that role divided by its number of units; because insertions are included, it can exceed 100%. We also report the *critical-word error rate* (CWER), the share of critical words (numbers, dates, names, places, course codes) not reproduced exactly; the *translation rate*, the share of embedded units replaced by their translation; and the *transliteration rate*, the share replaced by a sound-alike in the other script (§4).

Fragility and recovery. To ask whether quantization hurts embedded units more than matrix units, we compute on code-switched utterances

$$R = \frac{\Delta_{\mathrm{E}}}{\Delta_M}, \tag{2}$$

where Δ_{E} is the change in embedded-unit error (points) from the 16-bit to the quantized model and Δ_M the same change for matrix units ($\mathrm{M_s}$ and $\mathrm{M_i}$ together). $R > 1$ means quantization hurts embedded units more. R is reported only when the confidence interval of Δ_M excludes 0; otherwise the ratio is unstable and we report the two changes. Each quantized model is compared with a 16-bit run of the *same* backend, so that differences between decoders are not credited to quantization. For a repair applied to a quantized model, the recovery is

$$\mathrm{Rec} = \frac{E_q - E_{q,\mathrm{fix}}}{E_q - E_{16}}, \tag{3}$$

where E_{16}, E_q and $E_{q,\mathrm{fix}}$ are the embedded-unit errors of the 16-bit model, the quantized model and the quantized model with the repair. $\mathrm{Rec} = 1$ means the repair removes the whole quantization loss and 0 means it does

Set	Utts	Hours	CS%	E units	Spk
ASCEND train	9,869	8.8	28	7,081	18
ASCEND val	1,130	0.9	18	471	3
ASCEND test	1,315	0.9	28	772	2
NSC train	35,647	40.0	62	105,493	412
NSC val	3,367	3.0	25	4,099	60
NSC test (Same)	8,876	8.0	29	11,042	120
NSC test (Diff)	4,085	4.0	29	5,648	626
Stress (human)	120	0.18	100	227	3

Table 1: Data. CS% = share of utterances that contain a switch; Spk = speakers. NSC = NSC Part 4, Mandarin speakers; Same = Same Room, Diff = Different Room (Zoom). Stress = our phone recordings (test only).

not help. Rec is reported only where the quantization loss $E_q - E_{16}$ has a confidence interval excluding 0.

Confidence intervals. All intervals are 95% bootstrap intervals over 1,000 resamples, paired when comparing two systems on the same utterances. For NSC we also resample whole speakers (speaker-cluster bootstrap), which gives wider intervals; these are the ones we quote for NSC claims. ASCEND test has only two speakers, so its utterance-level intervals are optimistic.

4 Data

Corpora. ASCEND (Lovenia et al., 2022) is spontaneous Mandarin–English conversation; most of its sentences are Mandarin-main with English embedded. NSC Part 4 (Koh et al., 2019) is conversational Singapore speech; we keep the Mandarin-speaking participants, found from the speaker information sheet, so most sentences are English-main with Mandarin embedded. NSC comes in two recording conditions: *Same Room*, with both speakers in one room, and *Different Room*, recorded over Zoom. Table 1 gives the splits.

Sampling and splits. NSC training is capped at 40 h out of about 193 h of usable Same Room speech, taking code-switched utterances first (up to 30 h) and then English-only ones. NSC validation and test are random samples within a session-level speaker split, and the Different Room test is a random 4 h sample over all sessions. No Different Room test speaker appears in any training split; four speakers recorded in both rooms are removed from the Different Room test. ASCEND validation and test speakers do not appear in training. Code-switched NSC utterances contain more switches than ASCEND ones (2.85 vs 2.12 per utterance, Different Room) and longer embedded spans (2.8 vs 2.0 units), because NSC speakers were asked to switch.

Human stress recordings. To test critical words in real voices, we wrote 120 code-switched sentences dense in names, numbers, drug and course terms (*my* 密码 *has twelve characters now*), 60 English-main and 60 Mandarin-main. Three of the authors, English-speaking Singaporeans of Chinese background, each read a different 40 (20 of each kind), one take per sentence, into their own phone's built-in microphone in a quiet room at 20–30 cm, with about 1 s of silence at each end. The 48 kHz files (two readers AAC, one ALAC) go through the corpora's pipeline (16 kHz mono); clips last 3.8–7.2 s, 10.7 min in total. The readers confirmed that every clip follows its sentence exactly, so the prompt is the reference.

Labels. Word times come from forced alignment with MMS (Pratap et al., 2024) on romanised text (Hermjakob et al., 2018). On FLEURS (Conneau et al., 2023) clips spliced at known switch times, the aligned switch is a median of 10 ms from the true one. To tell translations from transliterations, two LLM annotators labelled 1,768 ASCEND reference–output pairs and their labels were adjudicated (Cohen's $\kappa = 0.84$ for translation vs. the rest). A word pair counts as a *translation* if it is in CC-CEDICT (MDBG) or its LaBSE similarity is at least $\tau = 0.65$, chosen on these labels; it counts as a *transliteration* if it is in the other script, is not a translation, and its phonetic similarity (pinyin and Metaphone codes, compared by edit distance) is at least $\theta = 0.1$. The critical-word list combines a rule tagger with a list labelled the same way ($\kappa = 0.77$). Two small human audits of these labels are reported in Limitations.

5 Methods

(a) Role-weighted loss. Standard fine-tuning gives every output token the same

cross-entropy loss. We give tokens of embedded units and of matrix units at a switch extra weight:

$$\mathcal{L}_{\text{role}} = -\sum_{t} \big(1+\alpha\, e_t+\beta\, s_t\big) \log p_\phi(y_t \mid y_{<t}, x), \tag{4}$$

where y_t is the t-th reference token, x the audio, p_ϕ the model's probability, e_t and s_t are 1 if token t belongs to an embedded unit or to a matrix unit at a switch, and α, β are the extra weights ($\alpha = \beta = 0$ is ordinary fine-tuning). Each token's role is found from the byte span it covers in the byte-level BPE vocabulary, which is exact even when one Chinese character is split across two tokens.

(b) Semantic minimum-risk loss. Following minimum word error rate training (Prabhavalkar et al., 2018), we build a set $\mathcal{C}$ of candidate transcripts for each clip and lower their expected cost:

$$\mathcal{L}_{\text{sem}} = \sum_{y \in \mathcal{C}} \hat{p}_\phi(y \mid x)\, c(y), \tag{5}$$

where $\hat{p}_\phi(y \mid x)$ is the model's probability of candidate y, renormalised over $\mathcal{C}$, and the cost $c(y)$ counts its unit errors, except that a correct translation of a non-critical embedded word costs $\rho_0 < 1$ instead of 1; every inserted unit costs 1. Beam search alone returns near-identical candidates, so $\mathcal{C}$ pools beam-4 outputs under a forced `<|zh|>`, a forced `<|en|>` and a free language token, plus one candidate per embedded word in which that word is swapped for its dictionary translation. Candidates are regenerated with the current model on a sub-batch every four steps. The full training loss is $\mathcal{L} = \mathcal{L}_{\text{role}} + \lambda\, \mathcal{L}_{\text{sem}}$, where λ sets the weight of the semantic term. We call fine-tuning with this loss the *full system*.

(c) Language-ID head. A two-layer classifier on Whisper's frozen encoder predicts English, Mandarin or non-speech for every 20 ms frame. It is trained on ASCEND and NSC frames labelled by forced alignment, plus 6,000 FLEURS clips spliced from word-sized pieces (15.9 h), including splices of one language between two speakers so that a change of speaker is not taken as a change of language. Predictions are smoothed with a 5-frame median filter and language runs shorter than 200 ms are merged into their neighbours.

(d) Re-ranking and gating. For each clip we pool up to 12 candidates (beam 4 under each of the three language prefixes) and pick one by a weighted sum of three scores: Whisper's log-probability; an acoustic match score from the MMS CTC model on the romanised candidate; and the agreement between each word's language and the language-ID head's prediction over that word's frames. The weights are tuned on validation. The *on-device* variant drops MMS, which is larger than Whisper small, and takes word times from Whisper's cross-attention. Candidates whose gzip compression ratio exceeds 2.4 (Whisper's own repetition check) are removed first. In the *gated* version, a clip is re-ranked only when the language-ID head finds a run of the minority language of at least θ_{gate} ms; otherwise the greedy output is kept.

(e) Quantization repairs. *Mixed precision*: we quantize one Transformer block at a time to NF4, rank blocks by the KL divergence of the output distribution on embedded-word tokens, and keep the top k blocks at 16-bit. *Quantization-aware fine-tuning*: QLoRA (Dettmers et al., 2023) trains 16-bit low-rank adapters (rank 16, one epoch, learning rate 10^{-4}) on top of the NF4 model with the model's own training loss; we evaluate the adapters both as they are and merged into the weights and re-quantized for whisper.cpp.

(f) Transliteration penalty. To target transliteration directly, we add an extra cost to each candidate:

$$c_{\text{tl}}(y) = c(y) + \gamma\, N_{\text{tl}}(y), \tag{6}$$

where $c(y)$ is the semantic cost above and $N_{\text{tl}}(y)$ is the number of embedded reference units that y writes as a sound-alike in the other script (detector of §4); a transliterated embedded unit thus costs $1 + \gamma$ instead of 1, and $\gamma = 0$ gives the semantic loss back.

6 Experimental Setup

Model. Whisper (Radford et al., 2023) is a Transformer encoder–decoder. The encoder reads an 80-bin log-Mel spectrogram of a 30 s window (shorter clips are padded; all our clips are under 30 s, so no chunking is needed) through two convolution layers and a stack

Setting	Value
Optimiser	AdamW, $\beta = (0.9, 0.98)$, $\epsilon = 10^{-6}$, weight decay 0.01, gradient clip 1.0
Learning rate	tiny $3.75{\times}10^{-5}$, base $2.5{\times}10^{-5}$, small $1.25{\times}10^{-5}$; 500 warm-up steps, then linear decay to 0
Batch, epochs	32 utterances (small: 16×2 accumulation); 3 epochs; bf16; seed 425
Targets	main-language token first; at most 128 tokens
Checkpoint	lowest mean val MER, checked every 1,000 steps on 600 utterances per val set
Role loss	$\alpha = 0.5$, $\beta = 0.5$
Sem. loss	$\lambda = 0.1$, $\rho_0 = 0.5$; candidates every 4 steps on a sub-batch of 8 (small: 4)
LID head	2 layers on the frozen encoder; AdamW, one-cycle, peak lr 10^{-3}; 2 epochs; batch 32
Re-ranking	≤ 12 candidates (beam 4×3 language prefixes); gzip ratio ≤ 2.4; gate $\theta_{\text{gate}} = 200$ ms
Mixed prec.	blocks kept at 16-bit: ft tiny 2, full base 3 (chosen on val)
QLoRA	rank 16, scale 32, dropout 0.05 on all attention and MLP projections; lr 10^{-4}, 100 warm-up steps, 1 epoch
Thresholds	translation $\tau = 0.65$; transliteration $\theta = 0.1$
Decoding	greedy, ≤ 128 tokens; language token restricted to zh/en (HF, CTranslate2)

Table 2: Key hyper-parameters. Learning rates and batch sizes are per model; loss weights were tuned on base (Appendix A) and reused unchanged.

of self-attention blocks; the decoder generates byte-level BPE tokens, starting with a language token. We use three sizes: tiny (39M parameters, 4 encoder and 4 decoder layers, width 384), base (74M, 6+6, width 512) and small (244M, 12+12, width 768). Audio is resampled to 16 kHz mono, and references are normalised into units as in §3.

Models and training. We fine-tune Whisper tiny, base and small on ASCEND and NSC training data together, updating all weights (Table 2). The loss weights were tuned on base by grid search ($\alpha \in \{0.5, 1, 2\}$, $\beta \in \{0, 0.5\}$, then $\lambda \in \{0.1, 0.3\}$, $\rho_0 \in \{0.2, 0.5\}$) and used unchanged for small: $\alpha = \beta = 0.5$, $\lambda = 0.1$, $\rho_0 = 0.5$ (Appendix A). The transliteration penalty was tested on base in two rounds: $\gamma \in \{0.5, 1\}$ at $\lambda = 0.1$, then $\lambda \in \{0.4, 1\} \times \gamma \in \{0, 3\}$; each round included a $\gamma = 0$ retrain to measure run-to-run noise. The gate threshold was chosen from $\theta_{\text{gate}} \in \{0, 200, 400, 800, \infty\}$ ms; 200 ms won for every model. Every selection rule was written down before any test result was examined.

Quantization. On GPU we use bitsandbytes LLM.int8 (Dettmers et al., 2022) and NF4 on all linear layers, compared with the 16-bit Hugging Face model. On CPU we use CTranslate2 int8 (compared with CTranslate2 float32) and whisper.cpp Q8_0, Q5_0 and Q4_0 (compared with whisper.cpp F16). CPU rows use a fixed subset of the test sets: all of ASCEND test, 4,000 NSC Same Room and 2,000 Different Room utterances.

Decoding and hardware. Decoding is greedy with a 128-token cap unless stated. Hugging Face and CTranslate2 restrict the language token to Mandarin or English; whisper.cpp chooses among all languages. The main training runs used one NVIDIA L40S GPU. Speed is measured on a MacBook Pro (Apple M2 Pro, 4 threads, plugged in) on 100 test clips totalling 259 s.

7 Running the System

All code is in one repository (`https://github.com/nibess1/cs425_codeswitch`); Table 3 lists the commands. Each step reads and writes plain files, so steps can be rerun separately. *Pre-processing* turns each corpus into 16 kHz FLAC clips and a JSONL manifest with the raw and normalised reference, the speaker, the main language and the duration. *Training* reads a YAML config whose fields can be overridden on the command line (Table 2). *Testing* decodes the test sets with a 16-bit, simulated 8/4-bit (bitsandbytes) or exported CPU model (CTranslate2, whisper.cpp) and writes one JSONL line per utterance. *Scoring* (`score.py`) computes every metric of §3 from those files with bootstrap intervals; every number in this report comes from it.

8 RQ1: Role or Language?

Embedded units fail most. In almost every system and test set, embedded units have the highest error, followed by matrix units at a switch and then other matrix units (Figure 1). For off-the-shelf small, embedded error is 56.8% on ASCEND against 23.4% for

```
# 0. Environment: Python 3.11, one CUDA GPU (CPU formats need ctranslate2 and whisper.cpp)
micromamba create -n cs425 -c conda-forge python=3.11 pip ffmpeg && micromamba activate cs425
pip install -r requirements.txt && pip install -e . && pytest        # normaliser + scorer tests
# 1. Pre-processing: 16 kHz mono FLAC clips + JSONL manifests in data/<corpus>/manifests/
huggingface-cli download CAiRE/ASCEND --repo-type dataset --local-dir data/ascend/hf
python scripts/prepare_ascend.py                        # official splits
python scripts/fetch_nsc_dropbox.py fetch data/nsc_part4 -j 2 --only cs-chn && python scripts/prepare_nsc.py
python scripts/prepare_fleurs.py                        # spliced clips for the language-ID head
python scripts/prepare_stress.py                        # our recordings: data/stress/audio/<reader>/<id>.wav
python scripts/align.py ascend_train nsc_train ascend_val nsc_val    # MMS word times
# 2. Training (checkpoints in ckpt/<run>/best)
python scripts/train.py configs/ft_base.yaml            # plain fine-tuning; also ft_tiny.yaml, ft_small.yaml
python scripts/train.py configs/ft_base.yaml --run full_base --alpha 0.5 --beta 0.5 --lambda_sem 0.1 --rho0 0.5
python scripts/train_lid.py --encoder ckpt/ft_base/best --run lid_ft_base
python scripts/train.py configs/ft_base.yaml --run qat_full_base --model ckpt/full_base/best \
       --lora true --lr 1e-4 --epochs 1 --warmup 100
# 3. Testing (predictions in preds/<model>__<precision>__<method>.jsonl)
python scripts/decode.py --model ckpt/full_base/best --model_tag full_base --precision nf4_bnb --method full
python scripts/deploy_eval.py --model ckpt/full_base/best --tag full_base --method full \
       --formats ct2_f32 ct2_int8 ggml_f16 ggml_q8_0 ggml_q4_0
python scripts/sensitivity.py --model ckpt/full_base/best --tag full_base      # mixed-precision blocks
python scripts/rerank.py --model ckpt/full_base/best --tag full_base --lid ckpt/lid_ft_base/head.pt --loop_filter
# 4. Scoring and tables (results/tables/*.csv, results/figures/)
python scripts/score.py full_base__fp16__full full_base__nf4_bnb__full     # all metrics, 95% CIs
python scripts/score.py --fragility full_base__fp16__full full_base__nf4_bnb__full --set nsc_test --main en
python scripts/make_results.py && python scripts/export_tables.py && python scripts/make_figures.py
GPU=0 bash scripts/queue_h.sh && python scripts/make_human_tables.py       # human stress recordings
```

Table 3: Commands to pre-process the data, train, test and score (run from the repository root; GPU chosen with `CUDA_VISIBLE_DEVICES`). The `scripts/queue_*.sh` files chain these steps exactly as we ran them.

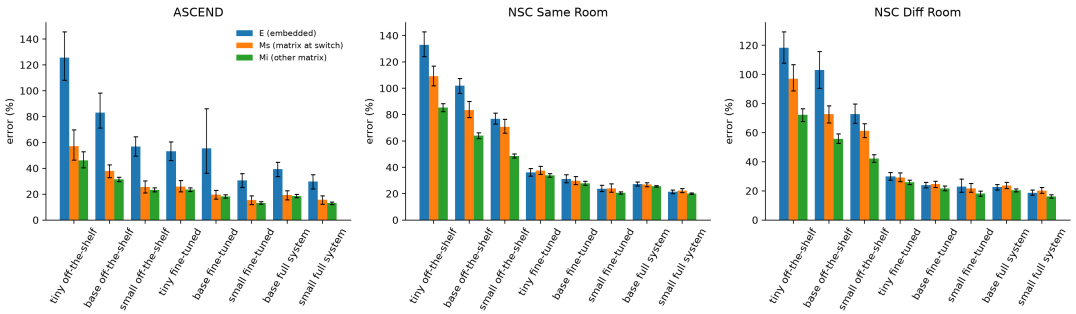


Figure 1: Error rate by role (E, M_s, M_i) for each system on the three test sets. Embedded units have the highest error in almost every system.

	Same word: E − matrix		Log-odds	
System	English	Chinese	b_{emb}	$b_{\text{emb}\times\text{en}}$
small, off-shelf	**+15.4**	**+26.0**	**0.94**	0.09
base, off-shelf	**+19.7**	**+32.0**	**1.18**	0.23
base, fine-tuned	**+8.0**	**+1.3**	0.01	**0.65**
small, fine-tuned	**+7.4**	+0.3	−0.07	**0.79**

Table 4: Role vs. language. Left: error (points) on the same word type when embedded minus when in the matrix language. Right: logistic-regression coefficients (§8). Bold: 95% CI excludes 0.

M_i, and 76.7% on NSC against 48.6%.

The same word is harder when embedded. Table 4 (left) compares each word type that occurs in both roles. For off-the-shelf small, an English word is 15.4 points [12.1, 18.6] more likely to be wrong when it is embedded than when it is part of an English-main sentence, and a Chinese character 26.0 points [23.9, 27.8] more. After fine-tuning, the gap for Chinese characters disappears (+0.3 for small) but the gap for English words remains (+7.4 [5.2, 10.1]).

To separate role from language, we fit a logistic regression on 46,916 units from 122 speakers:

$$\begin{aligned}\operatorname{logit} P(\text{err}) = b_0 &+ b_{\text{emb}}\,\text{emb} + b_{\text{en}}\,\text{en} \\ &+ b_{\text{emb}\times\text{en}}\,\text{emb}\cdot\text{en} \\ &+ b_f \log f + b_d\, d + b_c\,\text{nsc},\end{aligned} \tag{7}$$

where emb, en and nsc are 1 if the unit is embedded, is English, or comes from NSC; f is its training-data frequency and d its spoken duration; CIs come from a speaker bootstrap. Off the shelf, being embedded adds about one unit of log-odds over and above the unit's language

($b_{\text{emb}} = 0.94$ [0.78, 1.15] for small). After fine-tuning, b_{emb} is about zero, so embedded Mandarin is no longer penalised, but the interaction for embedded English stays large (0.79 [0.42, 1.11] for small). **Fine-tuning removes the role penalty for embedded Mandarin but not for embedded English.**

Errors are transliterations, not translations. We expected many embedded errors to be translations. They are not: only 0.5–1.9% of embedded units are replaced by a translation in the fine-tuned systems (3.4–4.0% off the shelf). Far more often, the embedded English word is written as Chinese characters that sound like it (deadline→得来, python→派森). This affects up to 10.9% of embedded units (off-the-shelf small on NSC 10.7%; full small 2.5–3.8%). The conclusion holds for every translation threshold $\tau \leq 0.7$ (Appendix A). A human audit of 30 random pairs confirms that translations are rare (none among the 26 labelled), but it labels most of the pairs our labels call transliterations as unrelated (§13), so the transliteration rates are best read as upper bounds.

9 RQ2: Methods at 16-bit

Fine-tuning matters most. Fine-tuning on ASCEND and NSC brings the largest gain, most of all on NSC, where off-the-shelf small reaches 55.1% MER and fine-tuned small 21.4% (Table 5). Forcing the wrong language token is harmful: off-the-shelf small with a forced `<|zh|>` token reaches 104.5% MER on NSC because it inserts text.

The losses help on NSC. Both losses lower embedded-unit error, and combining them works best (Table 6, top). The full system is significantly better than plain fine-tuning on both NSC test sets for base and small, in both MER and embedded error, and its critical-word error does not rise. On ASCEND the changes are mostly within noise (two test speakers, and a few long insertion runs widen the intervals). Because so few errors are translations (§8), MER with partial credit for translations is almost equal to MER, so the semantic loss acts mainly as a sequence-level minimum-risk loss.

A direct transliteration penalty does not help. Adding the penalty (f) to the full base system did not lower transliteration, MER or embedded error on any test set (Table 15 in the appendix). Its gradient was about 1/130 of the role-weighted loss's, because it acts only when a sampled candidate contains a transliteration; raised to match the main loss ($\lambda = 1$, $\gamma = 3$), it still did not beat the full system.

Re-ranking helps Mandarin-main speech, if it is gated. Re-ranking every clip lowers embedded error on Mandarin-main ASCEND (full base, on-device: −4.3 points) but raises MER on English-main NSC (full base, Different Room: +0.5; full small, Same Room: +0.8). Without the repetition filter, loops win the length-normalised score (plain fine-tuned base: 60.9% MER on NSC Same Room vs. 28.4% greedy), and the MMS score favours candidates that insert text. Gating fixes the problem (Table 6, bottom): re-ranking only the 25–28% of clips where the language-ID head finds a switch is never significantly worse than greedy decoding, and for full base it is significantly better on all three, while keeping most of the ASCEND embedded-error gain. The head labels 95–98% of speech frames correctly (Appendix A).

10 RQ3: Quantization

8-bit is safe. In every backend (LLM.int8, CTranslate2 int8, whisper.cpp Q8_0) and model size, no 8-bit setting hurts embedded units more than matrix units: no 8-bit R has an interval above 1, and the matrix change is mostly within noise.

4-bit hurts embedded units more in the smaller models. For fine-tuned tiny and base, 4-bit quantization raises embedded error 1.4 to 4.4 times as much as matrix error (Table 7; e.g. ft tiny NF4 on ASCEND: $R = 3.47$ [1.85, 6.13]). This happens for both embedded English (ASCEND) and embedded Mandarin (NSC), and in both NF4 and whisper.cpp Q4_0. On the corpora small is robust (but see §11): its errors change by about one point and R is about 1 or not reportable. Our off-the-shelf pilot had suggested the opposite ($R \approx 0.6$), but that does not carry over to fine-tuned models. In our per-block scan, the

	ASCEND		NSC Same Room		NSC Different Room	
System	MER	E	MER	E	MER	E
small, off-the-shelf	26.3	56.8	55.1	76.7	48.8	72.7
tiny, fine-tuned	26.1	53.2	34.5	36.2	26.8	29.8
base, fine-tuned	21.3	55.3	28.4	31.0	22.4	24.0
small, fine-tuned	14.8	30.6	21.4	23.8	19.0	22.9
base, full system	20.3	39.2	26.1	27.4	21.1	22.5
small, full system	**14.6**	**29.8**	**20.6**	**21.3**	**17.1**	**18.7**

Table 5: Test error (%) at 16-bit, greedy decoding. E: embedded-unit error. Full system = fine-tuning with $\mathcal{L}_{\text{role}} + \lambda\mathcal{L}_{\text{sem}}$. Speaker-cluster 95% CIs for small, full system: NSC Same Room MER [18.6, 23.4], Different Room [15.7, 18.6].

		ASCEND		NSC Same Room		NSC Different Room	
Compared with	System	MER	E	MER	E	MER	E
plain fine-tuning	base + role-weighted	−0.7	−14.6	−0.6	**−2.7**	−0.6	**−0.9**
	base + semantic only	−1.2	−14.6	**−1.3**	−1.7	**−1.1**	−0.4
	base, full system	−1.0	−16.1	**−2.3**	**−3.6**	**−1.3**	**−1.4**
	small + role-weighted	**−0.5**	−2.3	+0.1	−1.1	−0.9	−3.3
	small, full system	−0.1	−0.8	**−0.7**	**−2.4**	**−1.9**	**−4.2**
own greedy output	base full + gated re-ranking	**−0.7**	**−5.4**	**−0.4**	−0.4	**−0.3**	−0.7
	small full + gated re-ranking	**−0.5**	**−3.9**	−0.1	+0.2	+0.1	−0.1

Table 6: Paired differences in points (negative = better). Top: each method vs. plain fine-tuning of the same size. Bottom: on-device gated re-ranking vs. the same model's greedy output. Bold: 95% CI of the difference excludes 0.

Model	Format	Slice	Δ_{E}	Δ_M	R
ft tiny	NF4	A, zh	+16.2	+4.7	3.47
ft base	Q4_0	A, zh	+8.2	+1.9	4.40
full base	Q4_0	A, zh	+7.1	+2.3	3.07
full base	NF4	N, en	+3.0	+1.7	1.75
ft tiny	Q4_0	N, en	+8.9	+6.4	1.40

Table 7: 4-bit settings where R is reportable and its 95% CI excludes 1. Slice: A = ASCEND, N = NSC Same Room; zh / en = Mandarin-main (embedded English) / English-main (embedded Mandarin). Δ in points.

Repair	Model	Slice	Loss	Rec
Mixed (2 blocks)	ft tiny	A, zh	16.2	**0.65**
	ft tiny	N, en	11.2	**0.77**
Mixed (3 blocks)	full base	N, en	3.0	**0.70**
	full base	D, en	2.2	**0.48**
QLoRA (NF4)	ft tiny	A, zh	16.2	**0.83**
	ft tiny	N, en	11.2	**0.72**
	full base	N, en	3.0	**0.77**
	ft base	N, en	2.5	−0.30
QLoRA → Q4_0	ft tiny	N, en	8.9	−0.12
	full base	N, en	3.3	0.06

Table 8: Recovery of the 4-bit embedded-error loss (points). D = NSC Different Room. Bold: CI of Rec excludes 0. QLoRA → Q4_0: adapters merged and re-quantized for whisper.cpp.

first encoder layer is always the most sensitive block.

Repairs work for NF4 but not for the deployed format. Keeping the two or three most sensitive blocks at 16-bit, or adding QLoRA adapters, recovers about half to nine-tenths of the NF4 embedded-error loss for tiny and full base (Table 8); QLoRA does not help fine-tuned base, whose loss is small. When the QLoRA adapters are merged into the weights and the model is re-quantized to whisper.cpp Q4_0, the recovery disappears: no slice has a Rec interval above 0, and for tiny on the Different Room test the merged model is slightly worse than plain Q4_0. The adapters were trained against NF4's grid, not Q4_0's block-wise one, and whisper.cpp cannot load mixed-precision files at all. In practice the answer for whisper.cpp is Q8_0, which is as fast as Q4_0 on our laptop (§12) and changes embedded error by at most 1.2 points for any fine-tuned model.

11 Human Stress Recordings

Real voices are easier than synthetic ones, and RQ1 holds. Table 9 scores the phone recordings (§4) exactly as the corpora.

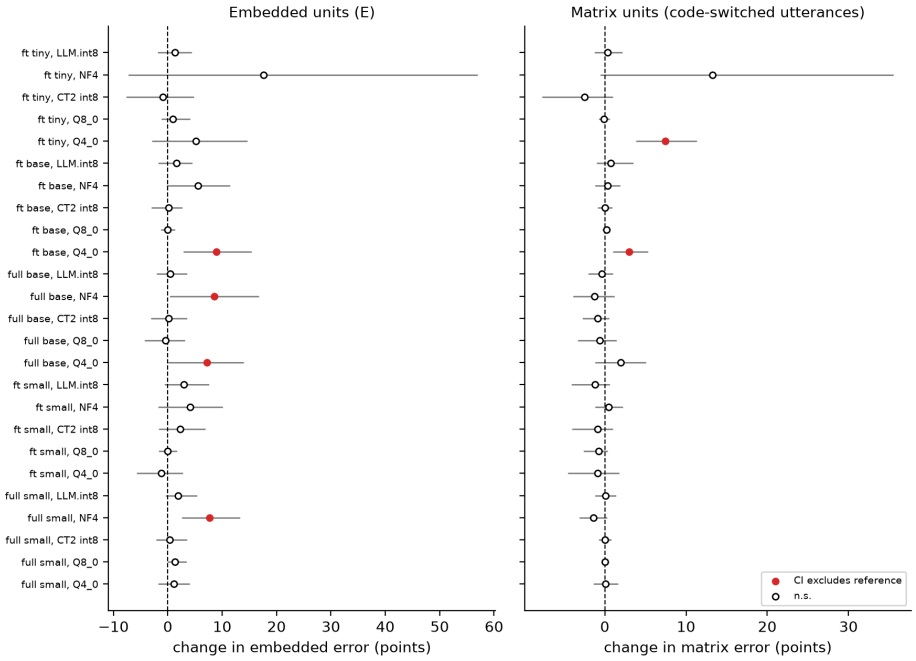


Figure 2: Human stress recordings: change in embedded (left) and matrix (right) error from each backend's 16-bit run to each quantized format, with 95% utterance-bootstrap CIs.

	16-bit		Δ_E		
Model	MER	E	Q8_0	NF4	Q4_0
small, off-shelf	33.6	83.9			
ft tiny	34.6	70.5	+0.9	+17.7	+5.1
ft base	23.0	49.7	0.0	+5.5	**+8.9**
full base	24.4	50.7	−0.4	**+8.6**	**+7.2**
ft small	14.3	31.4	−0.1	+4.1	−1.2
full small	13.4	27.8	+1.3	**+7.6**	+1.1

Table 9: Human stress recordings (120 utterances, 3 readers). Error (%) at 16-bit, and change in embedded error (points) to whisper.cpp Q8_0, NF4 and whisper.cpp Q4_0, each against its own backend's 16-bit run. Bold: 95% CI excludes 0 (utterance bootstrap; optimistic with 3 readers).

Fine-tuned models make 15–27 points fewer embedded errors here than on the same sentences in synthetic voices (full small 27.8% vs. 54.6%), which mispronounce embedded words. The role pattern is unchanged: off-the-shelf small gets 83.9% of embedded units wrong but only 23.5% of other matrix units, and writes 48% of embedded units as sound-alikes in the other script; fine-tuning cuts embedded error by 52.5 points [39.8, 65.3]. Matrix units next to a switch are harder than other matrix units in four of five fine-tuned models (full small: 12.4% vs. 9.0%). Embedded Mandarin is harder here than embedded English (full small 33.8% vs. 18.5%), but those sentences carry the rarer drug terms. With 227 embedded units, the full system and plain fine-tuning are within noise of each other (small, embedded: −3.6 [−10.0, 2.7]).

4-bit hurts embedded words even for small. 8-bit is again safe: no 8-bit setting in any backend changes embedded error by an interval that excludes 0 (Figure 2). At 4-bit, embedded error rises by 7–9 points for both base models (ft base Q4_0: +8.9 [2.9, 15.4]) and, unlike on the corpora, for full small in NF4 (+7.6 [2.6, 13.3]), while matrix error mostly does not change measurably (full small −1.4 [−3.1, 0.3]). So R is mostly not reportable; ft base Q4_0 gives $R = 3.0$ [0.9, 8.8]. whisper.cpp Q4_0 keeps small within 1.2 points. Tiny is dominated by repetition loops on single clips (without the worst NF4 clip its embedded change is +0.5), so we draw no conclusion for it.

Example outputs. Table 10 shows 16 recordings drawn at random. Off-the-shelf small keeps most embedded English when the sentence is Mandarin-main (insulin, module, release) but loses embedded Mandarin in English-main sentences, writing it as English sound-alikes (身份证→"shuffling jump",

贷款→“Taekwun”). Fine-tuned full small fixes most of these but still fails on rare terms (布洛芬→不落分, 会议室→“hui yi shah”, prednisolone, polyclinic→polytechnic) and once repeats a phrase instead of the sentence. In NF4 most outputs are unchanged; it adds errors on 身份证 (written 身分证) and 期末考, and happens to avoid the repeated phrase.

12 Deployment Cost

On the laptop (Table 11), whisper.cpp Q8_0 is as fast as Q4_0, as accurate as F16 and half its size: full small is a 252 MB file running at 0.57 times real time with 20.3% MER. The language-ID head raises base's real-time factor from 0.21 to 0.30 (CTranslate2) and re-ranking every clip to 1.76; gating, which re-ranks about a quarter of clips, gives an estimated 0.7.

13 Conclusion

Embedded words fail more often than the same words in the matrix language; fine-tuning closes this gap for embedded Mandarin but not English, and the remaining errors are rarely translations (many look like transliterations, though our automatic count is likely high). Role-weighted and semantic losses give small but reliable gains, and language-ID re-ranking helps only when gated. 8-bit quantization is safe; 4-bit hurts embedded words more than the rest, in tiny and base on the corpora and, on our phone recordings, even in small, and our repairs do not transfer to the deployed format. On device, 8-bit is the safe default.

Future work: more speakers and unprompted switching; losses that act on every embedded word against transliteration rather than on whole candidates; and repairs trained on the deployed format's own grid.

Limitations

Prompted switching. NSC speakers were asked to switch languages, and they switch more often within an utterance and in longer spans than ASCEND speakers (§4), so their switching may be more deliberate than natural code-switching.

Small ASCEND test set. ASCEND test has two speakers, so its utterance-level intervals are optimistic and several ASCEND differences are within noise. NSC claims use speaker-cluster intervals.

Automatic labels. The translation, transliteration and critical-word labels, and the check for partner speech leaking into NSC recordings, come from two LLM annotators with adjudication or from signal rules, not from people. Agreement is reported in Appendix A (translation vs. rest $\kappa = 0.84$; critical words $\kappa = 0.77$; leakage rule vs. annotator $\kappa = 0.29$). Two team members each audited a small sample, which we take as the final human check. On 26 random translation pairs, the human agreed with our labels on 58% [39, 77] ($\kappa = 0.15$ [0.00, 0.40]): neither side found a translation, but the human called 11 of the 13 pairs our labels call transliterations unrelated, so our transliteration rates are likely too high. On 50 critical-word candidates drawn from cases where the rule tagger and the LLM annotators disagreed, agreement with the adjudicated list was 66% [54, 78] ($\kappa = 0.32$ [0.12, 0.55]); the human was stricter, rejecting 15 of the 38 words the list marks critical. Both audits have one annotator and few items (95% bootstrap intervals over items), and the partner-leakage labels were not audited.

Small human stress set. The phone recordings (§11) are 120 read sentences from three readers, each reading different sentences once, so reader, phone and sentence are confounded, intervals are utterance bootstraps and optimistic, and most fragility ratios are not reportable. In 24 clips all fine-tuned models make the same error; besides shared misrecognitions (*opens*→*open*), these include homophones no recogniser can resolve (她/他) and normaliser mismatches (*level 3* vs. *three*) that slightly inflate every system's error. The earlier synthetic stress set (CosyVoice 2) is kept in the appendix for comparison only.

Scope of speed results. Speed was measured on one machine (Apple M2 Pro); the relative order of formats may differ on x86 or phone processors. Gated re-ranking speed is an estimate from its parts.

Reference (what was said)	Off-the-shelf small, 16-bit	Full small, 16-bit	Full small, NF4
我们约了 polyclinic 下午三点半	我們約了POLITICNIC下午3點半 [1]	我们约了 polytechnic 下午 3 点半 [1]	我们圆了 polytechnic 下午 3 点半 [1]
护士说 vaccine 打完要等十五分钟	護士說, vaccine 打完要等 15 分鐘 [0]	抢了 15 分钟抢了 15 分钟 [1]	忽视说 vaccine 打完要等 15 分钟 [0]
这个 python script 跑了四十分钟	这个排队script跑了40分钟 [2]	这个 python script 跑了 40 分钟 [0]	这个 python script 跑了 40 分钟 [0]
你记得带 insulin 去，放在冰箱第二层	你記得帶 Insulin 區放在冰箱第二重 [0]	你记得带 insulin 去放在冰箱第二从 [0]	你记得带 insulin 去放在冰箱第二从 [0]
这个 module 的 quiz 占百分之二十	這個Module的Quiz,占20% [0]	这个 module 的 quiz 占百分之二十 [0]	这个 module 的 quiz 占百分之二十 [0]
这个 bug 要在 release 之前 fix	这个bug要在release之前fix。 [0]	这个 bug 要在 release 之前 fix [0]	这个 bug 要在 release 之前 fix [0]
医生说 prednisolone 要慢慢减量	醫生說,Pretinicillin要慢慢解釋 [1]	医生说 pratnicillin your 慢慢减凉 [1]	医生说 pratnicillin your 慢慢减量 [1]
那个 report 要写三千个字	那个report要写3000个字 [0]	那个 report 要写 3000 个字 [0]	那个 report 要写 3000 个字 [0]
the 药房 closes at nine tonight	The yellow phone closes at 9pm tonight. [2]	the 药房 closes at nine tonight [0]	the 药房 closes at nine tonight [0]
the nurse said 布洛芬 only after meals	The nurse said bull often only after meals. [3]	the nurse said 不落分 only after meals [3]	dinner set 不落分 only after meals [3]
he forgot the 门卡 so he could not enter level five	He forgot the Munkah so he could not answer level 5. [2]	he forgot the 门卡 so he could not answer level five [0]	he forgot the 门卡 so he could not answer level five [0]
please bring your 身份证 for the exam	Please remember to bring your shuffling jump for the exam. [3]	please remember to bring your 身份证 for the exam [0]	please remember to bring your 身分证 for the exam [1]
my mother keeps the 药 in the second drawer	My mother keeps the yao in the second drawer. [1]	my mother keeps the 药 in the second drawer [0]	my mother keeps the 药 in the second drawer [0]
I booked the 会议室 from two to four	I bought the Huyish from 2 to 4. [3]	i book the hui yi shah from two to four [3]	i book the hui yi shah from two to four [3]
the 贷款 has twelve monthly payments	The Taekwun has 12 monthly payments. [2]	the 贷款 has twelve monthly payments [0]	the 贷款 has twelve monthly payments [0]
the 期末考 starts on the first of December	The team who caused that on the 1st of December [3]	the 齐木考 starts on the first of december [2]	the 起目靠 start on the first of december [3]

Table 10: Sixteen human stress recordings drawn at random (seed 425; top: Mandarin-main, bottom: English-main) and the raw outputs of three systems. Superscript: number of embedded units the system got wrong in that clip. Off-the-shelf Whisper often writes Traditional characters, which the normaliser maps to Simplified before scoring.

Model	Format	File	RSS	RTF	MER
ft tiny	Q8_0	42	135	0.086	31.6
full base	Q8_0	78	196	0.176	25.8
full base	Q4_0	44	161	0.172	28.7
full base	F16	141	261	0.336	25.7
full small	Q8_0	252	427	0.573	20.3
full small	Q4_0	139	315	0.584	21.0
full small	CT2 int8	238	839	0.742	21.1
full small	F16	465	650	1.299	20.3

Table 11: Cost on a MacBook Pro (M2 Pro, 4 threads). File and peak memory (RSS) in MB; RTF = processing time / audio time; MER (%) on the NSC Same Room CPU subset. Formats are whisper.cpp unless marked CT2 (CTranslate2).

Ethics and Data Licences

We use NSC Part 4 under the Singapore Open Data Licence version 1.0, which asks for the following notice: *Contains information from the National Speech Corpus (Part 4) accessed on 14 October 2026 from the Infocomm Media Development Authority of Singapore (IMDA), which is made available under the terms of the Singapore Open Data Licence version 1.0 (https://data.gov.sg/open-data-licence).* We use the data for research only and do not suggest that IMDA endorses this work. ASCEND and FLEURS are used under their published licences, and the synthetic stress-set voices were produced with CosyVoice 2 (Apache-2.0) from FLEURS prompts (CC-BY 4.0). The human stress recordings were made by three of the authors, who agreed to their use for this project; they are shared only with the course staff as part of the submission and are not released publicly. We collected no data from people outside the team. Speech recognition that runs on the de-

vice keeps audio off remote servers, which matters for the clinical and counselling uses that motivate this work; errors on names, numbers and drug terms in such settings can still cause harm, so outputs should be checked by a person where mistakes matter.

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A Additional Results

A.1 Hyperparameter selection

Grids were run on base and scored on pooled ASCEND + NSC validation (Table 12). Role weights: lowest embedded error among runs whose MER is at most 1 point above plain fine-tuning. Semantic loss: lowest semantic MER among runs whose critical-word error does not rise.

Run	MER	E	CWER	Chosen
plain fine-tuning	31.42	36.13	27.91	
α=0.5, β=0	31.57	35.04	28.35	
α=0.5, β=0.5	30.44	33.50	27.91	✓
α=1, β=0	34.34	44.82	27.47	
α=1, β=0.5	31.28	33.90	29.45	
α=2, β=0	32.49	33.69	28.57	
α=2, β=0.5	32.81	37.02	29.01	
λ=0.1, ρ_0=0.2	29.41	33.02	25.49	
λ=0.1, ρ_0=0.5	29.39	33.22	26.81	✓
λ=0.3, ρ_0=0.2	29.62	33.33	27.03	
λ=0.3, ρ_0=0.5	29.44	33.06	26.59	

Table 12: Validation grids on base (%). Semantic-loss runs use $\alpha = \beta = 0.5$.

A.2 Language-ID head

Head	Accuracy (%)	Boundary (ms)	Found (%)
shared (ft base)	96.0 / 95.4	60 / 120	75 / 85
own, full base	95.9 / 95.3	60 / 100	77 / 84
own, full small	97.7 / 96.4	20 / 80	78 / 85

Table 13: Language-ID head on validation (ASCEND / NSC). Accuracy on speech frames; median distance from a true switch to the nearest predicted one; share of switches detected.

The model's own head gives the same reranking accuracy as the shared head; its benefit is speed, because it reuses the model's encoder pass.

A.3 Label agreement and the translation threshold

Two LLM annotators labelled each item independently, the second seeing only the written rubric. On 1,768 ASCEND pairs, agreement was 0.798 (κ = 0.62) over three classes and 0.981 (κ = 0.84) for translation vs. the rest. On 335 candidate critical-word types, agreement was 0.899 (κ = 0.77); the rule tagger agreed less with either annotator (κ = 0.55 and 0.47), mostly because it has no named-entity component. Human audits of small samples agree less with these labels (κ = 0.15 for translation labels, n = 26; κ = 0.32 for critical words, n = 50; see Limitations); results table 20c lists every comparison with its interval. With τ between 0.6 and 0.7 the number of flagged translation occurrences changes by about ±20%, and translations remain a small share of embedded errors for every $\tau \leq 0.7$.

A.4 Partner speech and alignment

In NSC Same Room recordings the partner's voice is sometimes audible. A signal rule (overlap with the partner's transcript, speaker-embedding similarity, and output words that appear in the partner's transcript) flags 1.75% of Same Room and 0.15% of Different Room test utterances. Removing them lowers Same Room MER by about one point for every fine-tuned model (full small: 20.65 → 19.71) and changes no conclusion. Forced-alignment boundaries are within a median of 10 ms of known switches in FLEURS splices (87% within 50 ms) and a median of 73 ms from Whisper's cross-attention times at real switches.

A.5 Training on Different Room audio

NSC training data (40 h)	Same test	Diff (held-out)
Same Room (= ft base)	28.4	21.6
Different Room	32.3	23.6
20 h Same + 20 h Diff	28.3	22.2

Table 14: MER (%) of base when NSC training audio comes from the Different Room condition (Different Room split by session; ASCEND training data unchanged).

Training on Different Room audio does not help on Different Room test speech: training only on it is 2.0 points worse [0.4, 3.8] (Table 14). The gap between the two rooms in Table 5 is therefore not a mismatch between training and test conditions.

A.6 Transliteration penalty

At λ = 0.1, no γ passed the validation rule (MER at most 0.5 points and critical-word error no higher than the full system), so small was not trained. In the second round the

	MER			Transliteration rate		
Run	A	S	D	A	S	D
λ=0.1, γ=0 (control)	−0.14	−0.09	−0.08	+0.91	−0.07	+0.21
λ=0.1, γ=0.5	−0.27	+0.17	**+0.31**	+0.13	+0.05	−0.39
λ=0.1, γ=1	−0.37	−0.04	−0.02	−0.65	−0.25	−0.05
λ=1, γ=0 (control)	−0.12	−0.06	+0.16	+0.39	−0.31	−0.05
λ=1, γ=3 (chosen)	+0.12	+0.26	+0.19	+0.26	−0.04	−0.27

Table 15: Transliteration penalty on base: paired test differences vs. the full base system (points; negative = better). A = ASCEND, S = NSC Same Room, D = NSC Different Room. Bold: CI excludes 0. Embedded error changes are not significant for any penalised run.

rule allowed critical-word error up to 1 point higher, the size of the first round's retrain noise; only $\lambda = 1, \gamma = 3$ qualified, although it had the highest validation transliteration rate.

A.7 Synthetic stress set

We synthesised 120 code-switched sentences rich in critical words with CosyVoice 2 in four voices, clean and through a simulated phone channel (300–3400 Hz, 8 kHz, room reverberation, 20 dB noise). The phone channel roughly doubles errors and embedded words suffer most (full small: MER 22.3% clean, 36.8% phone; embedded error 54.6% and 74.5%). Embedded error is high even on clean speech: the same sentences read by people give 15–27 points lower embedded error for every fine-tuned model (§11), which confirms that the synthetic voices often mispronounce embedded words. This set therefore supports only relative comparisons.